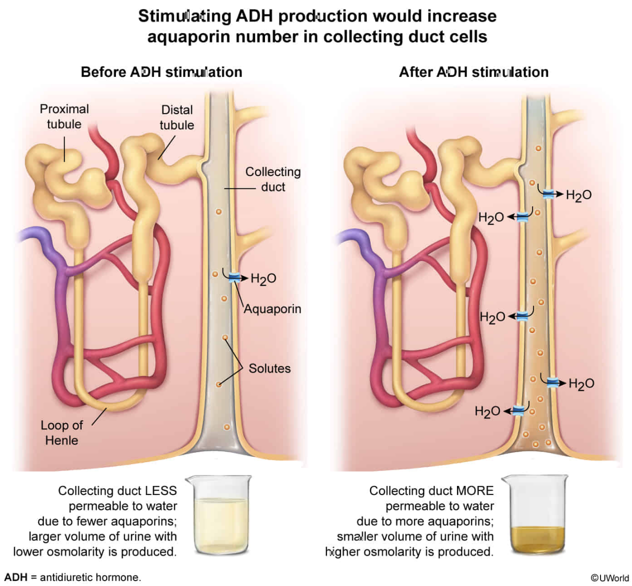
A method for increasing the quantity of aquaporins in the collecting duct cells of the kidney during a certain period would be:

- A. increasing extracellular sodium ion concentrations. [11%]
- B. stimulating aldosterone production. [14%]
- C. stimulating antidiuretic hormone production. [67%]
- D. increasing blood pressure. [6%]

Correct
67% answered correctly

Explanation:



Aquaporins are proteins found in cell membranes that facilitate the movement of water either into or out of the cell. As part of fluid homeostasis by the kidney, aquaporins can be inserted or removed from the apical membrane of collecting duct cells to regulate how much water is **reabsorbed** from the filtrate in the nephrons.

An increase in aquaporins in the collecting duct cells of the kidneys would increase the reabsorption of water in the nephrons, causing more water to remain in the body rather than in the urine. For example, if blood pressure were low, an increase in aquaporins would increase the amount of water reabsorbed into the blood. This would increase the blood volume and, in turn, the blood pressure. Conversely, if blood pressure were high, having fewer aquaporins in collecting duct cells would decrease the amount of water reabsorbed into the blood. This would decrease blood volume and thus decrease blood pressure, maintaining homeostasis **(Choice D)**.

The hormone responsible for activating a pathway that leads to the insertion of more aquaporins in collecting duct cells is antidiuretic hormone (ADH) or vasopressin. By **stimulating ADH production**, aquaporin insertion would

increase in collecting duct cells, causing an **increase** in water retained by the body.

(Choice A) An increase in extracellular sodium ion concentrations would *increase* the osmolarity (solute concentration) of the extracellular fluid, causing more water to remain in the body (ie, via osmosis) to maintain homeostasis. This would increase blood pressure, which would in turn inhibit ADH production, causing a *decrease* in the quantity of aquaporins in kidney collecting duct cells.

(Choice B) Aldosterone is a hormone stimulated by the renin-angiotensin system that acts on the distal tubules and collecting ducts of nephrons to promote the reabsorption of sodium ions and the secretion of potassium ions. Increased reabsorption of sodium ions increases the osmolarity (solute concentration) of the renal interstitial fluid. This elevated osmolarity promotes water *reabsorption*, which ultimately causes blood volume and blood pressure to increase. An increase in blood volume would inhibit the production of ADH, leading to a *decrease* in aquaporins in collecting duct cells.

Educational objective:
Antidiuretic hormone maintains fluid homeostasis when blood pressure is low by stimulating the insertion of aquaporins in the membrane of kidney collecting duct cells. This leads to increased water reabsorption, which raises the overall blood volume and blood pressure.

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